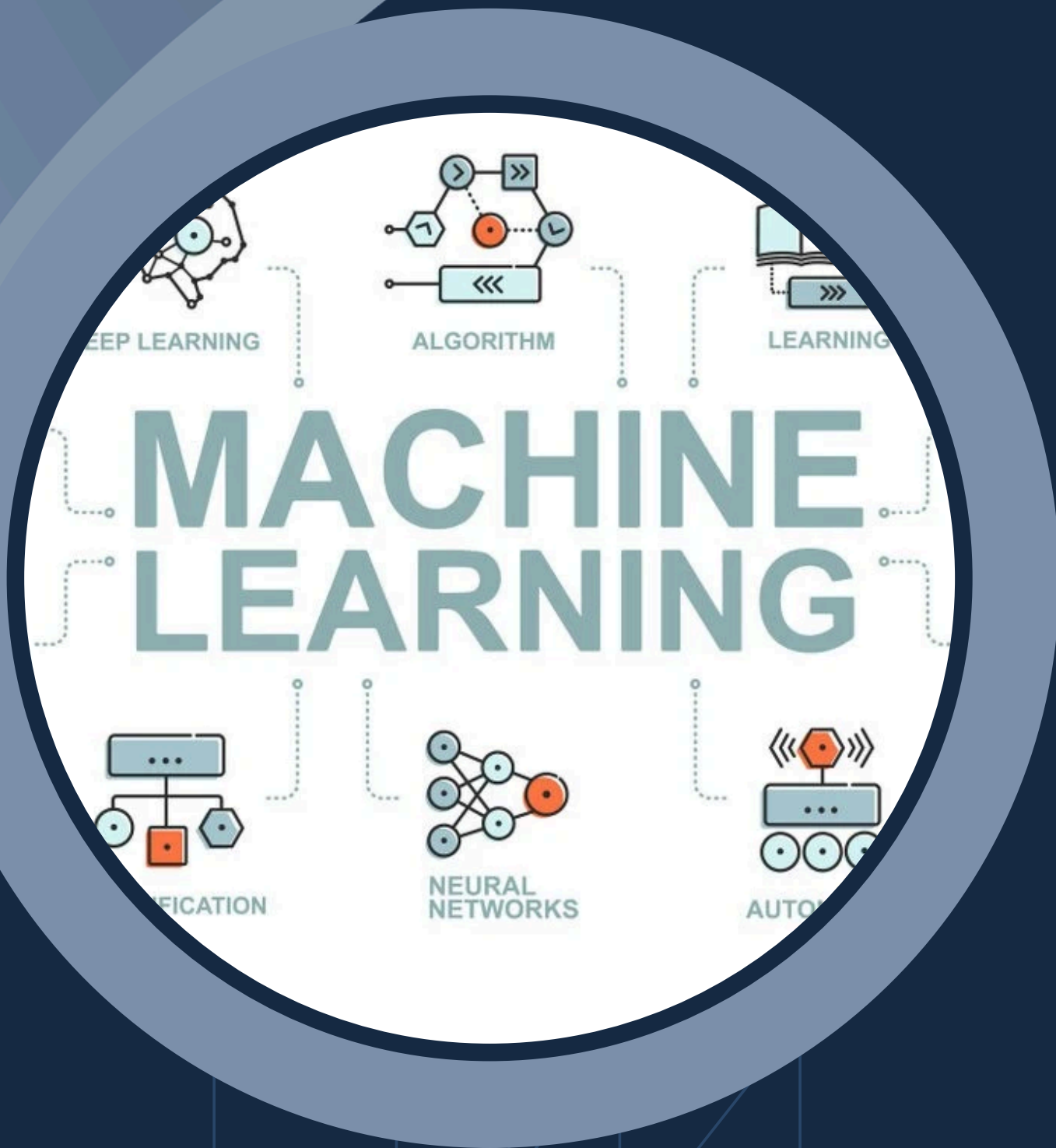




TAMWHEEL LITE

Tamweel Lite MACHINE LEARNING PROJECT

Predicting 90-Day Financing Default

PROBLEM STATEMENT

OBJECTIVE

Predict whether a financing applicant will default within 90 days, using a dataset of 10,000 synthetic applications.

BUSINESS CONSTRAINT

Review capacity is limited to 12% of all incoming applications — only the highest-risk cases can be flagged for manual review.

COST FUNCTION

Minimize total cost defined as:

$\text{Cost} = 10 \times \text{False Negatives} + \text{False Positives}$

Missed defaults (FN) are 10× more costly than false alarms (FP), reflecting real-world financial risk exposure.

SUCCESS METRIC

Optimize threshold and model selection to minimize total business cost under the 12% capacity constraint.

METHODOLOGY

01

Chronological split preserves temporal order; customer-aware split prevents data leakage across train and test sets.

90-day label maturity window ensures all financing outcomes are fully observed before labeling defaults.

02

03

Out-of-Fold (OOF) predictions generated on training data to enable unbiased model stacking and selection.

Pipeline: Data → OOF Predictions → Model Selection → Calibration → Decision Threshold.

04

RESULTS

Selected Model: Logistic Regression

OOF Average Precision: 0.39166

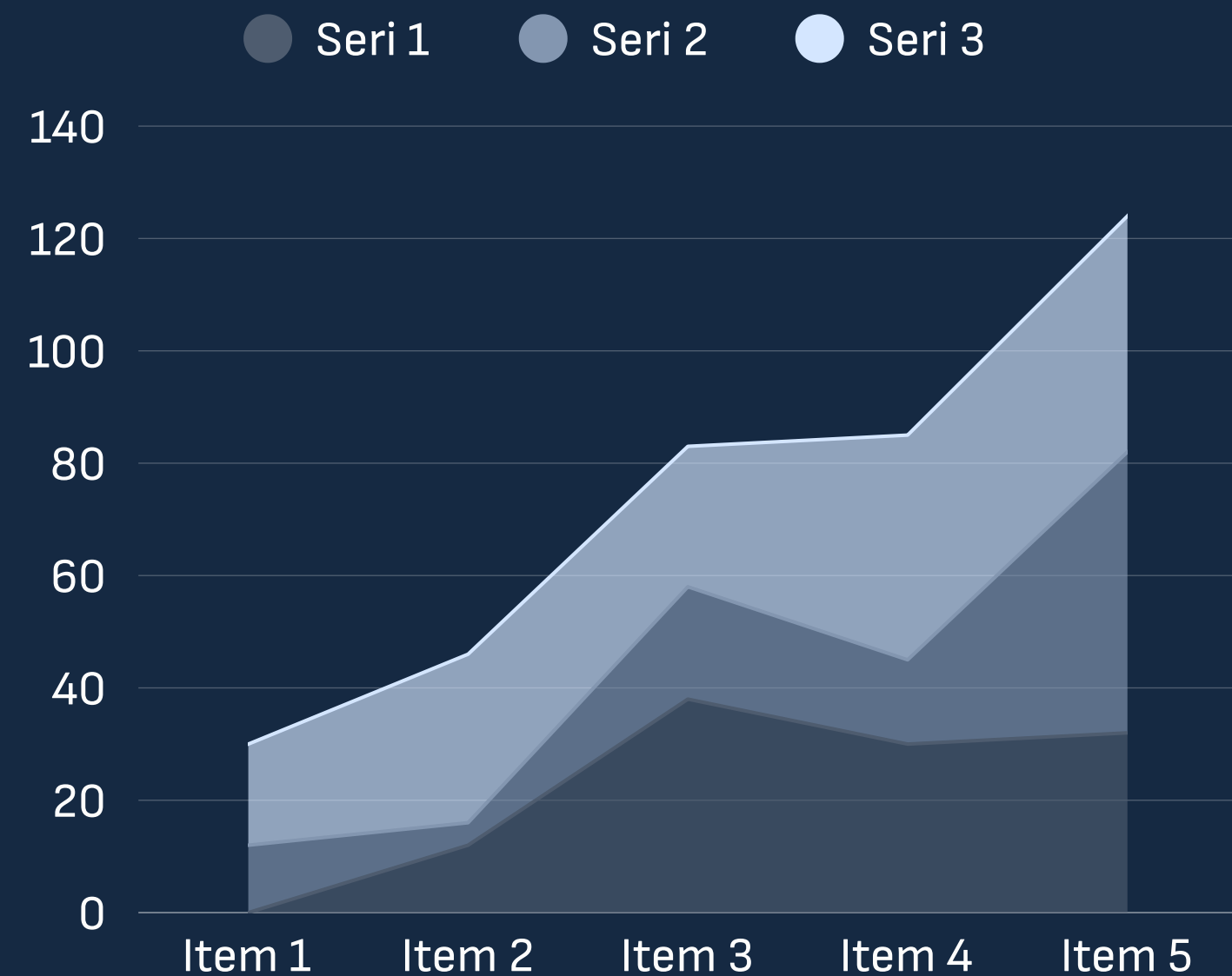
Model Comparison (OOF AP):

— Logistic Regression: 0.39166 ✓

— Weighted Ensemble: 0.38942

— Stacking: 0.38314

Logistic Regression outperformed all ensemble candidates on the held-out validation set. Its simplicity, calibration stability, and interpretability made it the optimal choice for production deployment.



FINAL DECISION

Operational Selection Summary

- Total applications evaluated: 2,500
- Applications above threshold (0.12225843): 330
- Final selections under 12% review capacity: 300
- Capacity constraint effectively binding — 30 flagged cases excluded
- Logistic Regression (OOF AP = 0.39166) drove the final threshold decision

Limitations & Next Steps

- Dataset is fully synthetic — real-world default patterns may differ significantly
- Challenge outcome is unknown; true model performance cannot be externally validated
- SHAP interpretations (Lab 04) reflect LightGBM, not the deployed Logistic Regression
- Sigmoid calibration slightly worsened Brier Score (0.07647 $\rightarrow$ 0.07806) — recalibration advised
- Independent evaluation on live data is required before production deployment